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Novel purine benzimidazoles as antimicrobial agents by regulating ROS generation and targeting clinically resistant *Staphylococcus aureus* DNA groove

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ABSTRACT

A novel series of purine benzimidazole hybrids were designed and synthesized for the first time with the aim to circumvent the increasing antibiotic resistance. Hexyl appended hybrid **3c** gave potent activities against most of the tested bacteria and fungi especially against multidrug-resistant strains *Staphylococcus aureus* (MIC = 4 µg/mL). Structure-activity relationships revealed that the benzimidazole fragment at the 9-position of purine played an important role in exerting potentially antibacterial activity. Both cell toxicity and ROS generation assays indicated that the purine derivative **3c** showed low cytotoxicity and could be used as a safe agent. Molecular modeling suggested that hybrid **3c** could bind with the residues of Topo IA through hydrogen bonds and electrostatic interactions. Quantum chemical studies were also performed on the target compound **3c** to understand the structural features essential for activity. The active molecule **3c** could effectively interact with *S. aureus* DNA to form **3c**-DNA complex through groove binding mode, which might block DNA replication to display their powerful antimicrobial activity.

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Microbial infection is a common and frequently occurring infectious disease worldwide. A large number of antibiotics and synthetic drugs have been widely used in clinic and have played an important role in the treatment of microbial infections.¹ However, the high morbidity and mortality caused by bacteria and fungi are still increasing serious threat due to the appearance of drug-resistant strains in recent decades.² Multiple strategies have been adopted to prevent and control antimicrobial resistance, one useful strategy is to develop the new drug hybrids which are generally considered a practical promise to tackle the growing problem of drug resistance.³

Purines can be regarded as one of the most ubiquitous and functional *N*-heterocyclic compounds found in nature. Purine scaffold is the core structural fragment of adenine and guanine in RNA and DNA.⁴ Purine nucleotides such as ATP, GTP, cAMP, cGMP, NAD and FAD act as co-factors, substrates or mediators in the function of many proteins.⁵ These proteins are estimated to include half

of the most pharmaceutically available targets, primarily enzymes and receptors. Some naturally existing and synthetic purines have been applied as antitumor, antiviral and antiparasitic agents and adenosine receptor ligands while the application in antimicrobial aspect has rarely been reported.⁶ Therefore, the development of purine-based novel antimicrobial agents is of significant value in medicinal chemistry. In view of the pivotal role of purines in the regulation of many biological processes, it arouses our immense interest to modify the purine skeleton using isosteric ring systems to develop novel structural molecules and investigate their possibility as new antimicrobial agents.

Azoles as nitrogen-containing five-membered aromatic heterocycles with desirable electron rich characteristic can readily bind with a variety of enzymes and receptors in biological systems through diverse weak noncovalent interactions, thereby exhibiting broad bioactivities.⁷ Thus, the combination of azoles and the purine nucleus provides a great prospect for medicinal and biological application. They have been developed as antimycobacterial and antiviral drugs and also as adenosine receptor agonists and antagonists. Especially, benzimidazole with special benzene-fused imidazole ring as a key component for various biological activities like antiparasitic, anticancer, anti-inflammatory and antiulcer agents etc. has attracted considerable concern.⁸ Recently, extensive

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biochemical and pharmacological studies revealed that benzimidazoles possessed large potentiality to inhibit the growth of bacterial and fungal strains.⁹ Further research disclosed that benzimidazole nucleus structurally resembling with purine scaffold in nucleotides was sometimes addressed as 1,3-dideaza-purine and had the capability of competing with purines, distinctly inhibiting the synthesis of nucleic acids and proteins, thereby killing bacterial strains or inhibiting their growth.^{10,11} Clearly, benzimidazole-based derivatives possess great potential as new antibacterial and antifungal agents. Reasonably, this structural similarity prompts us for the first time to combine benzimidazole moiety and purine backbone to generate a novel type of hybrids as potentially antimicrobial agents (Fig. 1) and evaluate their antimicrobial potency.

Design idea for target compounds was mainly from three aspects: (I) Structural modification on the benzimidazole nucleus; (II) Structural change on the purine scaffold; (III) The different linkers between benzimidazole and purine rings. The related considerations were given as follows:

- (1) The C-6 and N-9 positions in the purine scaffold are the most widely exploited anchoring points and the substituted purines are commercially available.¹² Therefore the purine nucleus through the N-9 position with benzimidazoles was hybridized;
- (2) The incorporated flexible $-\text{CH}_2-$ aliphatic linker between benzimidazole and purine nucleus is helpful for regulating the molecular conformation to interact with targets. The *N*-alkyl benzimidazole moiety was reported to have good performance in exerting bioactivities, the $-\text{CH}_2\text{CH}_2-$ aliphatic linker was also introduced into target compounds in order to evaluate its contribution towards antimicrobial efficacy;
- (3) A lot of work revealed that substituents at *N*-containing heterocycle could significantly influence the pharmacological properties by adjusting lipid-water partition coefficient and binding affinity. Rationally, various aliphatic chains with different lengths and substituted phenyl moieties including chloro and fluoro groups were introduced at N-1 position on benzimidazole moiety to investigate their effect on bioactivities;¹³
- (4) The substituted groups at C-2, 5, 6 positions on benzimidazole nucleus had the capacity to modulate the pharmacokinetic properties and enhance the antimicrobial efficacy. The substituents on benzimidazole nucleus were changed to explore their impacts on biological activities;
- (5) It is well known that saturated nitrogen heterocycles such as morpholine, piperidine and pyrrolidine are beneficial building blocks in drug design and prevalently present in many

approved drugs.¹⁴ The above mentioned alicyclic amines were introduced to C-6 position on purine nucleus for investigating the influence of the substituents in the purine nucleus on structure–activity relationship and bioactive profiles.

The designed structures of this series of novel purine benzimidazoles are shown in Schemes 1 and 2. All the newly synthesized compounds were characterized by spectral analyses and evaluated for their antimicrobial activities *in vitro* against five Gram-positive bacteria, six Gram-negative bacteria and five fungi according to the Clinical and Laboratory Standards Institute (CLSI).¹⁵ Structure-activity relationship, cytotoxicity, reactive oxygen species, molecular docking and quantum chemical studies as well as the preliminary antimicrobial mechanism with DNA to highly active compound were also discussed and investigated.

The target purine benzimidazole hybrids were prepared starting from commercial 6-chloro-9H-purine. Their synthetic routes were outlined in Schemes 1 and 2 and Schemes S1–S2. The reaction of 6-chloro-9H-purine with morpholine at 80 °C produced 4-(9H-purin-6-yl)morpholine **1** and then was further treated by benzimidazoles **8**, **10** and **12** in dimethylformamide to give the derivatives **2**, **3** and **4** respectively in yields of 48.8–71.5% (Scheme 1). The commercial *o*-phenylenediamines were reacted with various carboxylic acids through the cyclization to form benzimidazoles **7a–f**, which were subsequently proceeded *N*-alkylation with 1,2-dibromoethane in acetonitrile using potassium carbonate at 70 °C to afford *N*-substituted benzimidazoles **8a–f** in 46.3%–54.7% yields (Scheme S1). Intermediates **10** and **12** were prepared from the cyclization of *N*-mono substituted *o*-phenylenediamine with chloroacetic acid in hydrochloric acid at reflux in 74.2–83.6% yields. It was noticed that *N*-monosubstituted *o*-phenylenediamines **9** and **11** were provided in yields of 62.4–69.7% by the nucleophilic substitution reaction of *o*-phenylenediamine with a series of halogenated aliphatic chains at ratio of 1.2: 1 (Scheme S2). It was reported that the presence of aliphatic nitrogen heterocycles had some influence on the bioactivity, and thus target compounds **6a** and **6b** were further prepared to explore their effect on the bioactivities (Scheme 2). The starting material 6-chloro-9H-purine was also reacted with piperidine and pyrrolidine in ethanol at reflux using triethylamine as base to produce intermediates **5a** and **5b** with 92.7% and 91.4% yields, respectively. Further coupling with *N*-hexyl benzimidazole **10c** in dimethylformamide at 50 °C was carried out to generate the final products **6a** and **6b** in 67.4% and 69.3% yields.

The *in vitro* antimicrobial activities for all the target compounds were evaluated for five Gram-positive bacteria (Methicillin-Resistant *Staphylococcus aureus* N315 (MRSA), *Enterococcus faecalis*,

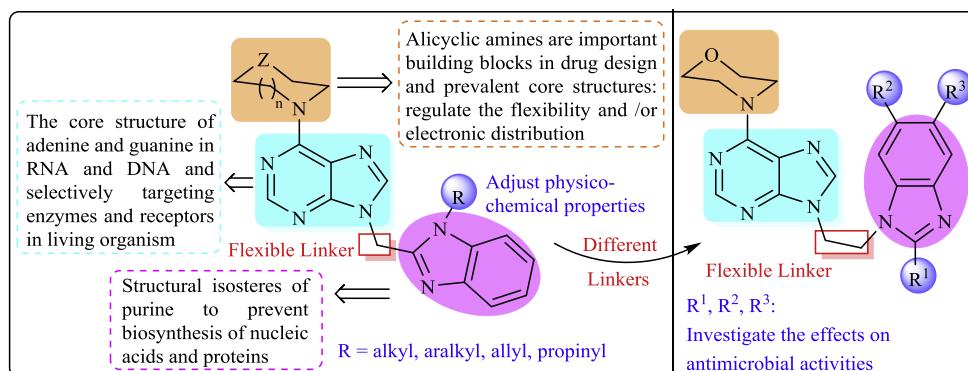
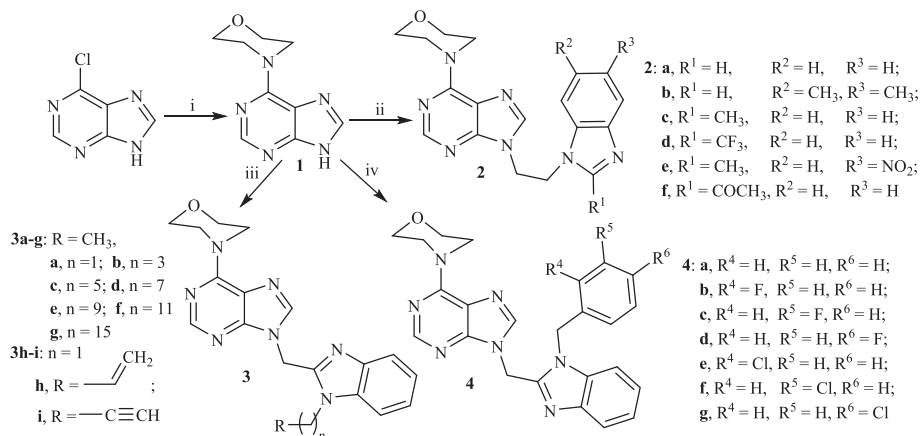
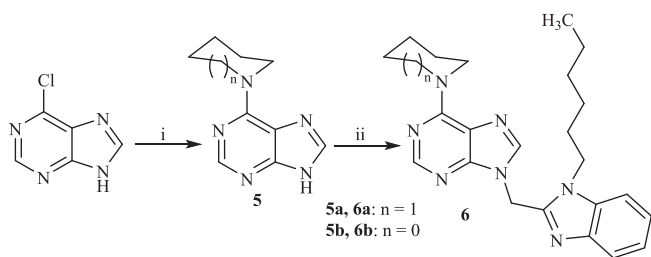


Fig. 1. Design of novel purine benzimidazole hybrids.



Scheme 1. Synthetic routes of purine benzimidazoles **2–4**. Reagents and conditions: (i) morpholine, 80 °C; (ii) benzimidazoles **8a–f**, potassium carbonate, dimethylformamide, 50 °C; (iii) benzimidazoles **10a–i**, potassium carbonate, dimethylformamide, r.t.; (iv) benzimidazoles **12a–g**, potassium carbonate, dimethylformamide, r.t..



Scheme 2. Synthetic routes of purine benzimidazoles **6a–b**. Reagents and conditions: (i) piperidine/pyrrolidine, triethylamine, ethanol, reflux; (ii) compound **10c**, dimethylformamide, potassium carbonate, 50 °C.

Staphylococcus aureus, *Staphylococcus aureus* ATCC 25923 and *Staphylococcus aureus* ATCC 29213), six Gram-negative bacteria (*Klebsiella pneumoniae*, *Escherichia coli*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa* ATCC 27853, *Escherichia coli* ATCC 25922) and five fungi (*Candida albicans*, *Candida tropicalis*, *Aspergillus fumigatus*, *Candida albicans* ATCC 90023, *Candida parapsilosis* ATCC 22019) using two folds serial dilution technique in 96-well micro-test plates recommended by CLSI with the positive control of clinically antimicrobial drugs norfloxacin (Table 1) and fluconazole (Table S1). As shown in Table 1, most of the prepared compounds gave weak to moderate antibacterial activity against the tested strains *in vitro*. Preliminary active screening revealed that morpholine derivative **3c** showed better antibacterial efficacy than other alicyclic amine ones **6a** and **6b**, and more importantly hexyl-appended hybrid **3c** exhibited stronger inhibition ability against most of the tested bacterial strains among the all target compounds.

Further studies focused on the influence of alkyl chains with different lengths (series **3**) and aromatic frameworks with different halogen moieties (series **4**) on antimicrobial activity (Table 1). It could be observed that the antibacterial efficacies of series **3** and **4** were comparable or even better than compound **1** towards some tested strains like *S. aureus* and *P. aeruginosa* ATCC27853 manifesting the importance of benzimidazole moieties. It was indicated that long alkyl chains (over ten carbons) should be not beneficial to regulate the hydrophilicity of this purine benzimidazole system for their antibacterial activity. The hexyl and decyl modified derivatives **3c** and **3e** exhibited relatively stronger antibacterial activities than reference norfloxacin against *P. aeruginosa* ATCC27853. It was noteworthy that hybrid **3c**, with prominent MIC value of 4 µg/mL against *S. aureus*, 16-fold more potent than clinical drug norfloxacin, had immense potentiality to be

developed as anti *S. aureus* agent. However, the allyl and propargyl derivatives **3h–i** did not display obvious effects on suppressing the growth of bacteria, which indicated that the unsaturated aliphatic chain was not propitious to exert inhibitory potency.

Among phenyl ones, fluorobenzyl derivatives **4b**, **4d** and **4f** also gave equivalent or even lower inhibitory concentrations than chlorobenzyl ones **4c**, **4e** and **4g** towards *S. aureus* ATCC29213, *P. aeruginosa* and *P. aeruginosa* ATCC27853, which implied that fluorine atom on the benzyl moieties exerted positive effect on microbial inhibition. Probably because fluorine atom could easily and efficiently form non-covalent forces thereby being helpful for the biological transportation and distribution in organism. In addition, it was also found that the substituted position also exerted some influence on the antibacterial activity with the order of the *o*-fluoro/chlorobenzyl derivatives > *m*-fluoro/chloro ones > *p*-fluoro/chloro ones against most of the tested strains.

The introduction of benzimidazoles at the 9-position of purine scaffolds using $-\text{CH}_2\text{CH}_2-$ as a bridge yielded compound series **2**. None of the tested strains was sensitive to compound **2a**.

However, the further modifications at the benzimidazole nucleus showed large effect on antibacterial activity. The electron-withdrawing trifluoromethyl, nitro and acetyl groups and electron-donating methyl groups were introduced at the 2-, 5- and 6-positions of the benzimidazole moiety. It was found that the electron-withdrawing groups were effective for enhancing anti-Gram-negative bacteria activities. Specifically, 2-acetyl derivative **2f** showed good MIC values (8 µg/mL) against *P. aeruginosa* and stronger antibacterial potency than reference drug against *K. pneumoniae* (32 µg/mL) and *P. aeruginosa* ATCC27853 (32 µg/mL). In comparison to electron-withdrawing groups, the electron-donating methyl purine benzimidazoles **2b–c** showed much poorer inhibitory potency.

The antifungal evaluation *in vitro* displayed that some prepared purine benzimidazoles exhibited moderate bioactivities against the tested fungal strains (Table S1). Compound **1** showed poor antifungal activities with MIC values ranging from 256 to 512 µg/mL, while compounds **3c**, **6a** and **6b** exerted stronger antifungal potency which suggested that alicyclic amines and benzimidazole moieties were positive to antimicrobial activities.

In the series of purine benzimidazoles **3a–i**, the lengths of aliphatic chains exhibited obvious effects on antifungal activity. The target molecule **3c** exerted the relatively good activities in inhibiting the growth of all the tested fungal strains, especially possessed lower MIC values (32–128 µg/mL) than the reference fluconazole (64–256 µg/mL) against clinically drug-resistant strains. Among benzyl series of purine benzimidazoles **4a–g**, 2- and 3-fluorobenzyl

Table 1
Antibacterial data as MIC ($\mu\text{g/mL}$) for target compounds **1–6**.^{a,b,c}

Compds	Gram-positive bacteria					Gram-negative bacteria					
	MRSA	<i>E. faecalis</i>	<i>S. aureus</i>	<i>S. aureus</i> ATCC25923	<i>S. aureus</i> ATCC29213	<i>K. pneumoniae</i>	<i>E. coli</i>	<i>P. aeruginosa</i>	<i>A. baumannii</i>	<i>P. aeruginosa</i> ATCC27853	<i>E. coli</i> ATCC25922
1	512	128	256	512	512	128	256	256	256	256	256
2a	256	128	256	256	256	256	256	128	256	256	128
2b	256	128	256	256	128	64	256	32	256	128	256
2c	512	256	256	32	128	256	512	256	256	256	256
2d	256	64	32	64	128	64	32	128	64	64	128
2e	256	128	16	128	64	128	128	64	32	64	16
2f	256	64	32	64	128	32	128	8	64	32	128
3a	256	128	256	256	256	256	128	128	256	256	128
3b	512	256	256	256	128	256	64	256	256	128	128
3c	16	128	4	64	512	64	128	64	64	16	8
3d	512	256	512	512	256	256	256	256	256	128	256
3e	256	128	256	256	256	128	32	256	256	16	128
3f	512	128	16	512	512	256	32	128	512	128	256
3g	512	32	64	256	256	128	512	512	256	256	256
3h	512	128	256	256	128	128	256	256	128	128	256
3i	512	256	128	128	256	128	128	256	256	128	512
4a	256	128	512	512	256	128	256	256	256	128	256
4b	256	64	256	128	16	64	64	32	128	32	256
4c	512	256	256	256	32	64	128	128	256	64	256
4d	512	128	256	32	64	16	256	256	256	128	256
4e	512	128	256	128	128	128	256	512	256	256	256
4f	256	256	512	64	256	128	128	256	256	64	256
4g	512	256	512	128	256	256	256	512	512	128	256
5a	512	256	32	512	512	256	256	256	128	64	256
6a	64	128	16	64	512	64	256	128	64	128	16
5b	512	512	64	512	512	512	256	256	256	128	256
6b	64	256	32	64	512	128	256	128	64	256	16
A	>512	256	64	32	8	>512	512	4	32	512	1

^a Minimum inhibitory concentrations were determined by micro broth dilution method for microdilution plates and MICs was completely inhibited concentration.

^b MRSA, Methicillin-Resistant *Staphylococcus aureus* N315; *E. faecalis*, *Enterococcus faecalis*; *S. aureus*, *Staphylococcus aureus*; *S. aureus* ATCC25923, *Staphylococcus aureus* ATCC25923; *S. aureus* ATCC29213, *Staphylococcus aureus* ATCC29213; *K. pneumoniae*, *Klebsiella pneumoniae*; *E. coli*, *Escherichia coli*; *P. aeruginosa*, *Pseudomonas aeruginosa*; *A. baumannii*, *Acinetobacter baumannii*; *P. aeruginosa* ATCC27853, *Pseudomonas aeruginosa* ATCC27853; *E. coli* ATCC25922, *Escherichia coli* ATCC25922.

^c A = Norfloxacin.

ones resulted in relatively good activities against *C. albicans* ATCC90023 with the MIC values of 64 and 32 $\mu\text{g/mL}$, respectively. The replacement of fluorobenzyl moiety by chlorobenzyl fragment did not give obvious effects on antifungal activities which displayed that fluorine atom was advantageous in inhibiting the growth of microorganisms. The change of linker $-\text{CH}_2-$ to $-\text{CH}_2-\text{CH}_2-$ yielded compounds **2a–f**, exhibited low activities against all test strains. Notably, compound **2e** showed much stronger activities (64 $\mu\text{g/mL}$) against *C. tropicalis* than fluconazole (MIC = 256 $\mu\text{g/mL}$).

Computational methods can be used to predict important pharmacokinetic properties of drug candidates and have been employed to rationalize biological activity. The molecular electrostatic potential (MEP) surface gives an indication of the charged surface area and hydrophilicity of compounds.¹⁶ Therefore, MEP maps of the target compounds were produced to investigate key structural features such as steric and electrostatic interactions, hydrogen donor/acceptor properties, and lipophilicity. As shown in Table 2, the selection of compounds at the neutral state resulted in a comparison of MEP surfaces by DFT-B3LYP theoretical calculations.

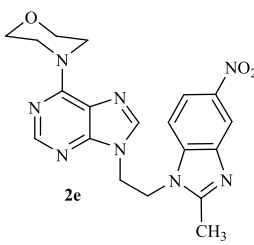
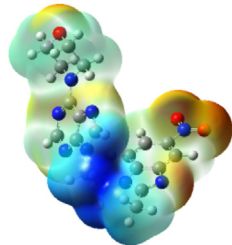
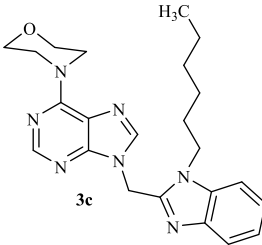
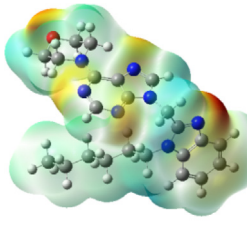
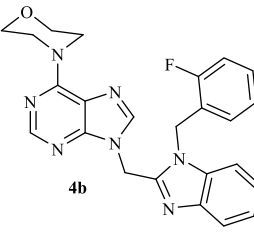
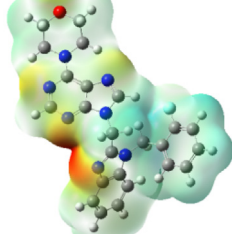
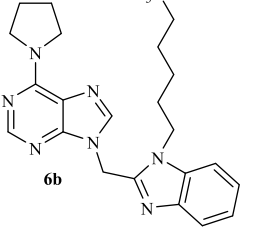
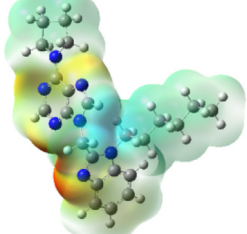
In the comparison to the MEP maps of the two modes of purine benzimidazoles, series **3** and **4** were taken as examples. The most noticeable difference came from the substituents at the 1-position of the benzimidazole nucleus. In the optimized structures of series **3** showing folded shape, oxygen and nitrogen atoms with electron negatively properties had a strong ability to interact with the enzyme. Because of the π - π conjugative effect and steric hindrance, series **4** with unfolded shape possessed weaker negatively charged surface area. Compared to the MEP surfaces of target compounds **3**, the electron negatively charged zones of the nitrogen

atoms were slightly decreased as a result of the replacement of morpholine by the pyrroline at the 6-position of the purine ring. When using $-\text{CH}_2\text{CH}_2-$ group as bridge, the electron positively charged zones greatly enhanced. On the other hand, the introduction of the electro-withdrawing groups like nitro group had much influence on the electron negatively properties of the molecules.

The active molecule **3c** was further evaluated for its toxicity *in vitro* against human lung adenocarcinoma cells (A549) by the colorimetric cell proliferation MTT assay. Experimental results revealed that the cell viability of A549 cell against compound **3c** was more than 85%, which suggested that this molecule showed low toxicity to the cells at concentrations <500 $\mu\text{g/mL}$ and the cell viability slowly decreased with the increasing concentration of compound (Fig. 2). Compound **3c** did not affect the cell viability of A549 within the MIC value. All demonstrated that compound **3c** possessed low cell toxicity which was considered to be one of the most important properties for active drug candidates.

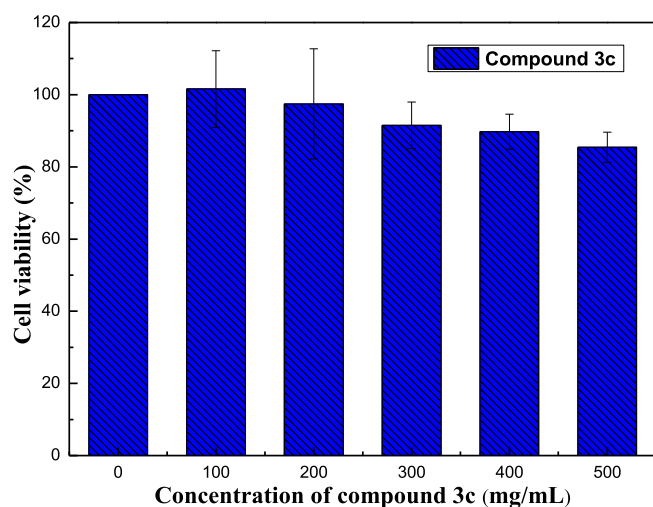
Evading persistent drug resistance in bacteria is essential for the recovery of human health and impels intervention strategies. *E. coli* and other life threatening infectious pathogens such as *S. aureus* and *M. tuberculosis* are significantly sensitive to changes in the intracellular oxidative environment. ROS are metabolic byproducts of aerobic respiration and are responsible for maintaining redox homeostasis in cells.¹⁷ Therefore, small molecules that enhance intracellular ROS can interfere with the cellular oxidative environment and induce cell death, and hence can serve as a new standard for determining cytotoxicity. ROS level was investigated using DCFH-DA as a fluorescence probe and imaged by fluorescence microscope. As shown in Fig. 3, target compound **3c** showed much lower ROS generation than H_2O_2 at concentration of 6.25, 12.5, 25, 50 μM , respectively. Therefore, the newly synthesized compound

Table 2Comparison of the MEP surfaces of target compounds **2e**, **3c**, **4b** and **6b**.

Compds	MEPs	Compds	MEPs
 2e		 3c	
 4b		 6b	

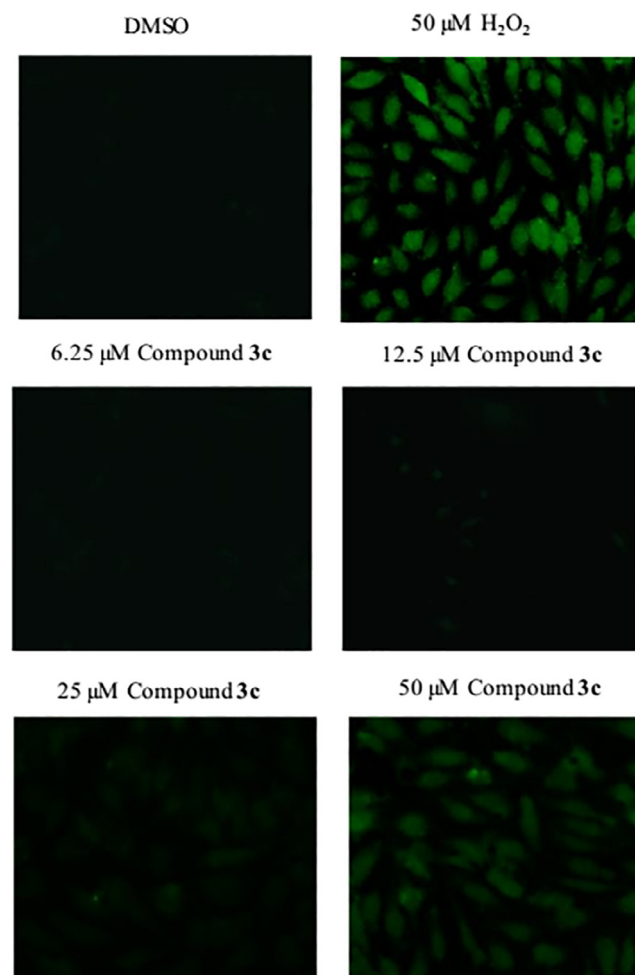
-8.000e-2

8.000e-2

**Fig. 2.** Cytotoxicity assay with compound **3c** in the A549 cell line were tested by MTT methodology. Each data bar is an average of five replicates.

3c could possess an excellent safety profile by regulating ROS generation.

To rationalize the observed antimicrobial activity and to understand the possible mechanism of the hybrids, molecular docking of active compound **3c** with DNA topoisomerase IA (Topo IA) was undertaken. The crystal structure data was obtained from the Protein Data Bank. The interaction of compound **3c** with Topo IA receptor was shown in Figs. 4 and S5. The docking result of interaction between compound **3c** and Topo IA might rationalize the possibly antibacterial mechanism. The nitrogen atoms of benzimidazole ring at 3- position and purine ring at 3-position were in close vicinity to the residue GLN 197 of Topo IA through hydrogen bonds with distance of 2.8 and 2.0 Å, respectively, which revealed that the nitrogen atom played important role in biological activity. Besides, electrostatic interactions existed between the aromatic fragment of compound **3c** and residues THR 496, VAL 196 and ILE 500 in Topo IA. Specifically, morpholine ring could also form electrostatic interactions with residue TYR 177. These hydrogen

**Fig. 3.** Compound **3c** induced ROS formation in HeLa cells and H₂O₂ as positive control.

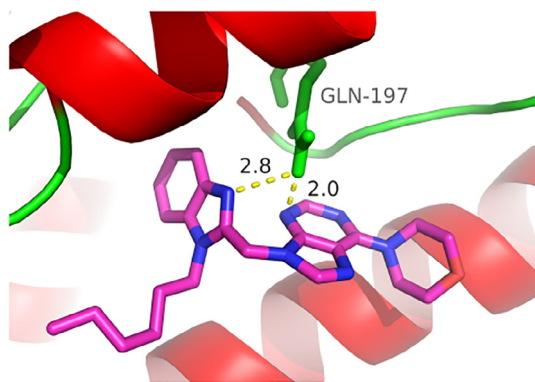


Fig. 4. Three-dimensional conformation of compound **3c** docked in Topo IA (PDB code: 3PWT).

bonds and electrostatic interactions might be responsible for the good inhibitory efficacy of compound **3c** against the tested strains. Furthermore, quantum chemical studies were also carried out to validate the capability of hydrogen bond formation through the nitrogen atoms of purine skeleton and benzimidazole nucleus (Tables 2 and S3).

To investigate the preliminary mechanism of antimicrobial action, the binding studies *in vitro* between the active molecule **3c** and DNA isolated from *S. aureus* bacterial strain (Figs. S6 and S7) on molecular level were carried out by UV–vis spectroscopy. With a fixed concentration of DNA, UV–vis absorption spectra were detected with the increasing concentration of compound **3c**. As shown in Fig. S9, UV–vis spectra displayed that the maximum absorption peak of DNA at 260 nm was gradually enhanced, simultaneously along with a slightly red shift during titration with the proportional addition of compound **3c** which was different from compound **3c** alone (Fig. S8). Meanwhile, spectral data displayed that the absorption value of simply sum of free DNA and free compound **3c** was a little greater than the measured value of **3c**–DNA complex which revealed a weak hypochromic effect existed between DNA and compound **3c**. This hypochromic effect on the spectra of **3c**–DNA complex and the slight red spectral shift might be resulted from the charge attraction of prepared molecule **3c** to DNA base eventually leading to formation of binary complexes. Moreover, the intercalation of the aromatic fragment of compound **3c** into the helix and the strong overlap of π – π^* states in the large π –conjugated system with the electronic states of DNA bases were in accordance with the observed spectral changes.

To learn more about the interaction between compound **3c** and DNA, the absorption spectra of competitive interaction of compound **3c** was investigated. Neutral Red (NR) is a planar phenazine dye and has been widely employed as probe due to its lower toxicity, higher stability and more convenient application than other common probes. Furthermore, spectrophotometric and electrochemical techniques have already sufficiently demonstrated that NR can bind with DNA through an intercalative mode. Therefore, NR was taken as a spectral probe to study the binding mode of **3c** with DNA in our work. The absorption spectra of the NR dye upon the addition of DNA are shown in (Fig. S11). Experimental results displayed that the absorption peak of the NR around 460 nm gradually decreased with increasing concentration of DNA and a new band at around 530 nm developed, which suggested the formation of new DNA–NR complex. An isosbestic point at 504 nm provided evidence of DNA–NR complex formation. Fig. S12 displayed the absorption spectra of a competitive binding between NR and **3c** with DNA. As shown, with the increasing concentration of **3c**, an apparent intensity change was not observed in the developing band around 460 nm. Compared with the

absorption around 460 nm of free NR in the presence of the increasing concentrations of DNA (Fig. S11), the absorbance at the same wavelength did not exhibit the reverse process (inset of Fig. S12). The results disclosed that compound **3c** failed to intercalate into the double helix of DNA by substituting for NR in the DNA–NR complex.

The fluorescence quenching spectra of compound **3c** with increasing amounts of DNA at the excitation wavelength of 278 nm was shown in Fig. S13. The fluorescence intensity of compound **3c** at around 316 nm regularly decreased, but the maximum emission wavelength of compound **3c** did not obviously change with the increase of DNA concentration. The results indicated that DNA could quench the intrinsic fluorescence of compound **3c** and the binding of compound **3c** with DNA indeed existed.

Dynamic quenching and static quenching are the two well-known quenching processes. Dynamic quenching is a process that the fluorophore contact with the quencher during the transient presence of the excited state, whereas static quenching results in the formation of fluorophore and quencher complex. The method to distinguish static and dynamic quenching is to calculate their different binding constants based on temperature or viscosity or to measure their lifetimes. In this work, the calculation of binding constant dependent on the temperature was used to elucidate the quenching mechanism.

The fluorescence quenching data can be analyzed by the well-known Stern–Volmer equation:

$$\frac{F_0}{F} = 1 + K_{SV}[Q] \quad (1)$$

where F_0 and F represent fluorescence intensity of compound **3c** in the absence and presence of DNA, respectively. K_{SV} (L/mol) is the Stern–Volmer quenching constant, and $[Q]$ is the concentration of DNA. The Stern–Volmer equation was applied to determine K_{SV} by linear regression of a plot of F_0/F against $[Q]$. Hence, the value of K_{SV} at different temperatures could be calculated and was shown in Fig. S14.

The results showed that in dynamic quenching, the quenching constants were expected to increase with a gradually increasing temperature due to larger diffusion coefficients. While the increase of temperature is likely to lead to a smaller K_{SV} in static quenching due to the dissociation of weakly bound complexes. Fig. S14 showed that the Stern–Volmer quenching constant, K_{SV} , decreased linearly with increasing temperatures. These calculated K_{SV} were 1.87×10^4 ($R = 0.9999$), 1.52×10^4 ($R = 0.9999$) and 1.21×10^4 ($R = 0.9999$) L/mol at 294, 304 and 314 K, respectively, at pH 7.4. Thus, the fluorescence quenching mechanism of compound **3c** interacted with DNA was verified to be resulted from static quenching. Therefore, the fluorescence quenching of compound **3c** by DNA could be described by the modified Stern–Volmer equation:

$$\frac{F_0}{\Delta F} = \frac{1}{f_a K_a} \frac{1}{[Q]} + \frac{1}{f_a} \quad (2)$$

where ΔF is the difference in fluorescence intensity of compound **3c** in the absence and presence of DNA at concentration $[Q]$, f_a is the fraction of accessible fluorescence, and K_a is the effective quenching constant for the accessible fluorophores, which are analogous to associative binding constants for the quencher–acceptor system. The dependence of $F_0/\Delta F$ on the reciprocal value of quencher concentration $1/[Q]$ is linear with the slope equaling to the value of $1/(f_a K_a)$. The value of $1/f_a$ is fixed on the ordinate. The constant K_a is a quotient of the ordinate $1/f_a$ and the slope $1/(f_a K_a)$. The modified Stern–Volmer plots were showed in Fig. S15 and the calculated results were depicted in Table S4. The decreasing trend of K_a with increasing temperature was in accordance with K_{SV} 's as discussed above, which accorded with static quenching.

Generally, the noncovalent interaction such as hydrogen bonds, van der Waals force, hydrophobic force and electrostatic interactions are the principal force for binding between small molecules and biomacromolecules. The main evidence to confirm the binding force is the thermodynamic parameters of binding reaction. If the enthalpy change (ΔH) does not vary significantly over the studied temperatures range, then its value and that of entropy change (ΔS) can be determined from the van't Hoff equation:

$$\ln K = -\frac{\Delta H}{RT} + \frac{\Delta S}{R} \quad (3)$$

where K is analogous to the associative binding constants at the corresponding temperature and R is the gas constant. The temperatures used were 294, 304 and 314 K, respectively. The values of ΔH and ΔS were obtained from the slope and intercept of the linear van't Hoff plot based on $\ln K_a$ versus $1/T$ (Fig. S16). The free energy change (ΔG) was then calculated from the following equation:

$$\Delta G = \Delta H - T\Delta S \quad (4)$$

Table S5 listed the thermodynamic parameters for the interaction of compound **3c** with DNA. The negative values of ΔG indicated that the binding process was spontaneous. The negative values of ΔH suggested that the binding process was primarily driven by the enthalpy and through the hydrogen binding interactions. The negative ΔS value is often used as an evidence for vander Waals force. Therefore, $\Delta H < 0$ and $\Delta S < 0$ obtained in this case manifested that both hydrogen bonds and vander Waals force exerted an important role in the binding of compound **3c** to DNA and was conducive to the stability of the complex.

It is well known that the highly negatively charged quencher is expected to be repelled by the negatively charged phosphate backbone of DNA. Therefore, if a molecule is protected from being quenched by anionic quencher, the binding pattern of the small molecule with DNA is intercalation. While different from intercalation, groove binding provides much less protection for the bound molecule. The value of quenching constants (K_{SV}) for the groove binding should be higher than that of the molecule bounding to DNA by intercalative pattern. Negatively charged I^- ion was chosen for this purpose. The values of K_{SV} of compound **3c** by I^- ion in the absence and presence of DNA were calculated to be 28.74 ($R = 0.994$) and 78.24 ($R = 0.998$) L/mol, respectively, by the Stern-Volmer equation (Fig. S17). The results showed that iodide quenching effect was enhanced when compound **3c** was bound to DNA, suggesting that the binding mode of compound **3c** with DNA might be of groove binding nature.

In conclusion, we have successfully developed a series of novel purine benzimidazole hybrids as safely antimicrobial agents via convenient and efficient synthetic procedures. All new compounds were confirmed by 1H NMR, ^{13}C NMR, IR and HRMS spectra. Some of the synthesized compounds exhibited good or even superior bioactivities to reference drugs against the tested bacterial strains especially against the clinically resistant *S. aureus*. The structure-activity relationship found that the benzimidazole moiety and various substituents played important roles in the biological activity and hexyl derivative **3c** was the most prominent one (MIC = 4 μ g/mL against *S. aureus*). Moreover, the active molecule **3c** also displayed low cell toxicity and could be used a safe agent. Molecular docking revealed that compound **3c** could bind with the residues of Topo IA through hydrogen bonds and electrostatic interactions. Further quantum chemical studies validated the capability of hydrogen bond formation through the nitrogen atoms of purine skeleton and benzimidazole nucleus which was consistent with the binding mode obtained from the docking result. The preliminary action mechanism investigation revealed that hybrid **3c** could interact with DNA isolated from *S. aureus* through groove binding driven by hydrogen bonds and van der Waals force to form

3c-DNA complex, thus exerting its powerful antibacterial and antifungal activities. Hence, compound **3c** could open new prospect for being developed as therapeutic agent for the purpose of coping multidrug resistant bacterial infections. Further researches, including the *in vivo* bioactive evaluation, resistance study, time-kill kinetic assay and introduction of various functional substituents into the hybrid backbone to investigate the effects on antimicrobial activities are now in progress in our group.

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A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.bmcl.2018.03.046>.

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